

DSCI 552 - Homework 2

Yang Zhao
USC NetID: yzhao528

Combined Cycle Power Plant Regression and ISLR Exercises

Data note: the supplied workbook contains five shuffled versions of the same data. As required, all analysis below uses Sheet1. The workbook headers AT and PE correspond to the assignment notation T and EP, respectively.

1. Combined Cycle Power Plant Data Set

(a) Data

The supplied Sheet1 contains the full Combined Cycle Power Plant data used here. No rows were dropped and no values were imputed.

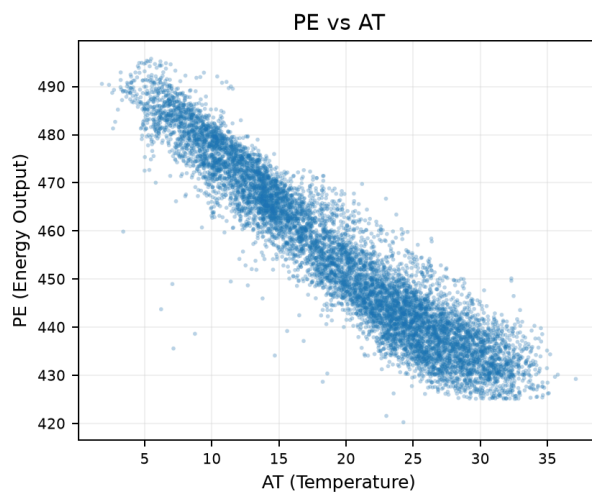
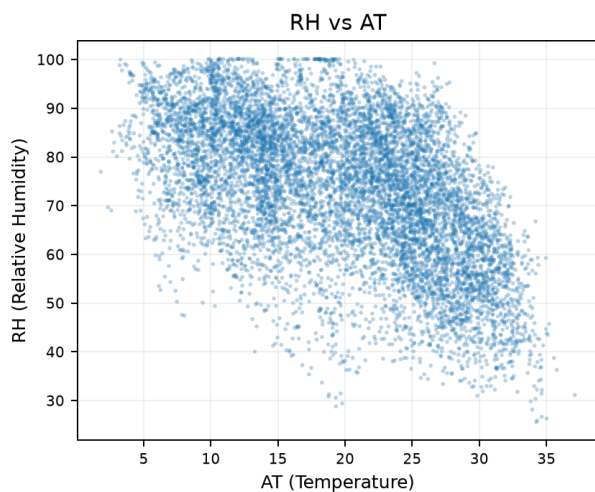
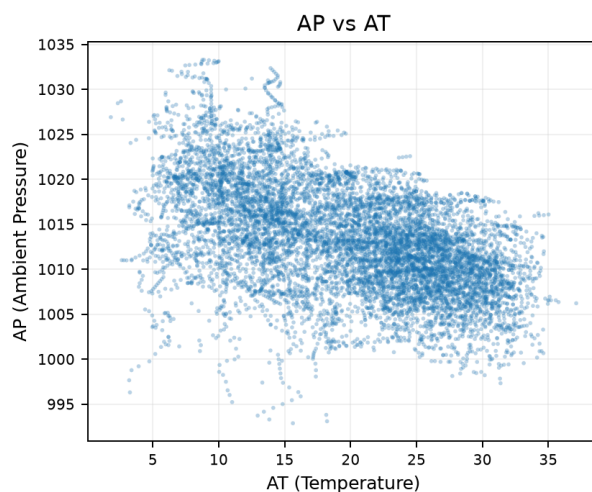
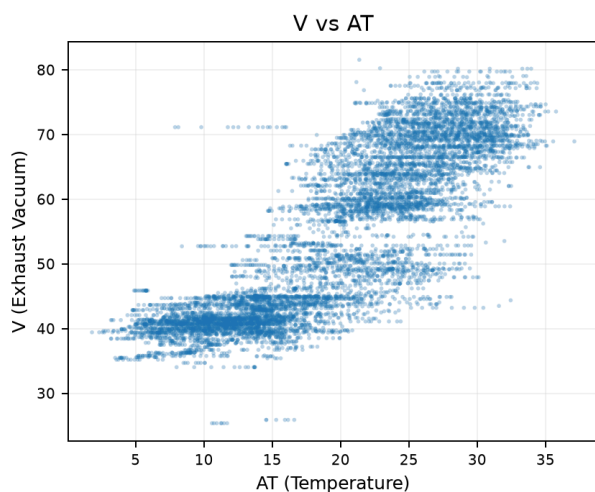
(b) Exploring the data

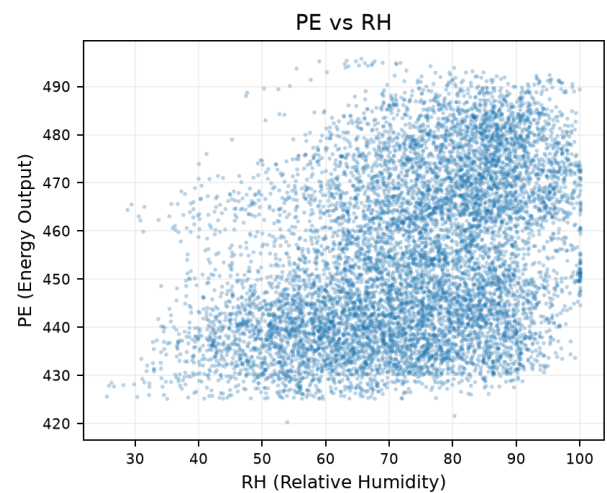
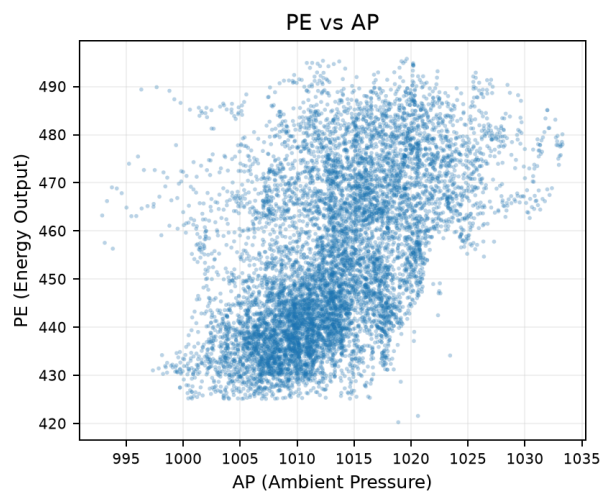
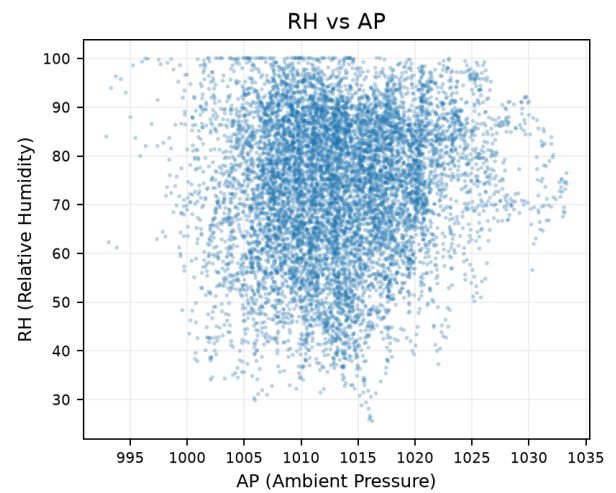
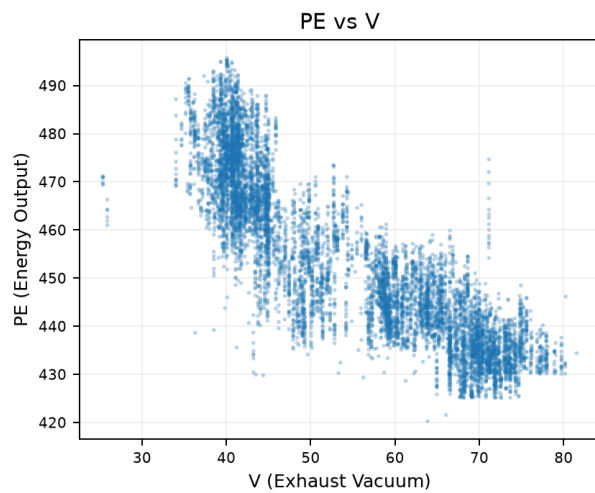
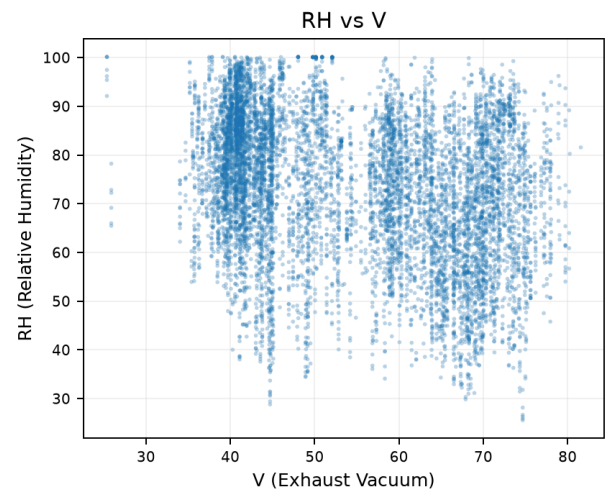
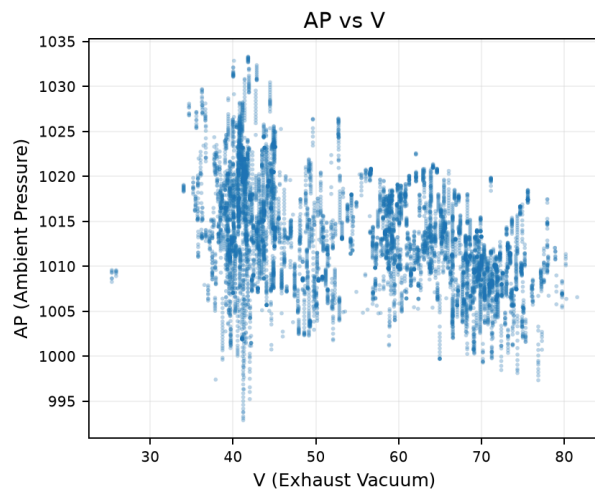
(b)(i) Rows, columns, and meaning

There are 9,568 rows and 5 columns. Each row is one hourly observation collected while the plant operated at full load. The columns are AT (ambient temperature), V (exhaust vacuum), AP (ambient pressure), RH (relative humidity), and PE (net hourly electrical energy output). The first four are predictors; PE is the response.

(b)(ii) Pairwise scatterplots and findings

The strongest relationship with PE is the negative association with AT ($r = -0.948$), followed by V ($r = -0.870$). AP has a moderate positive marginal association with PE ($r = 0.518$), and RH a weaker positive marginal association ($r = 0.390$). Among predictors, AT and V are strongly positively correlated ($r = 0.844$), while AT is negatively associated with AP and RH. These predictor-predictor relationships foreshadow the coefficient changes between simple and multiple regression.





(b)(iii) Descriptive statistics

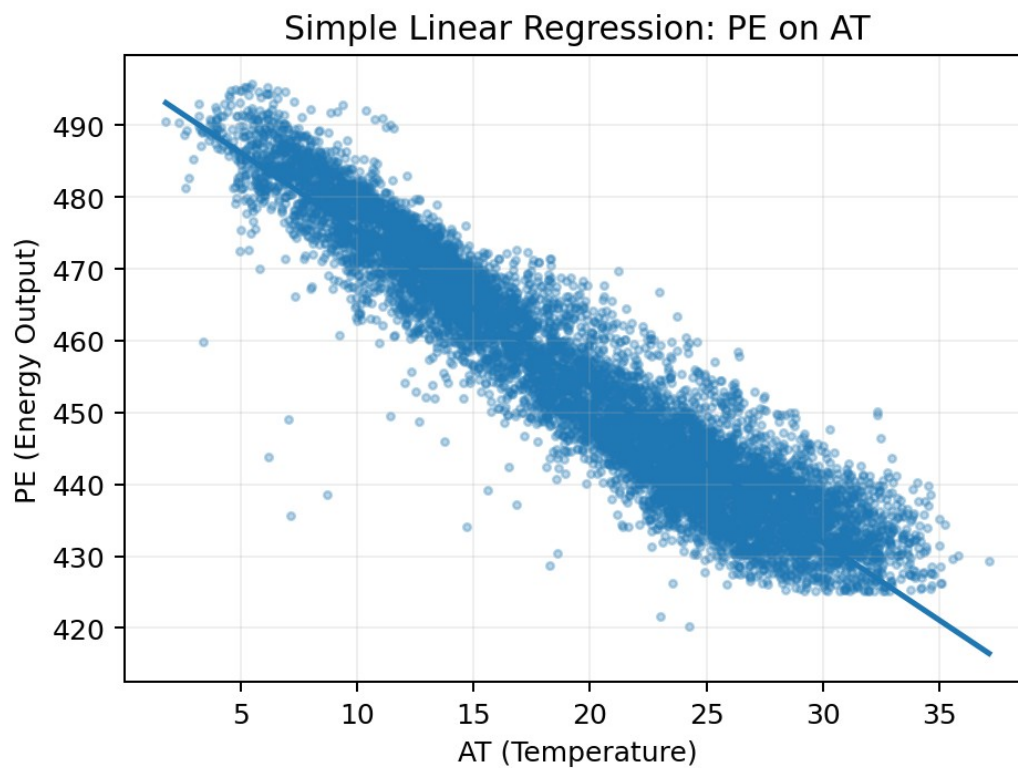
Variable	Mean	Median	Range	Q1	Q3	IQR
AT	19.651	20.345	35.300	13.510	25.720	12.210
V	54.306	52.080	56.200	41.740	66.540	24.800
AP	1013.259	1012.940	40.410	1009.100	1017.260	8.160
RH	73.309	74.975	74.600	63.328	84.830	21.502
PE	454.365	451.550	75.500	439.750	468.430	28.680

(c) Simple linear regressions

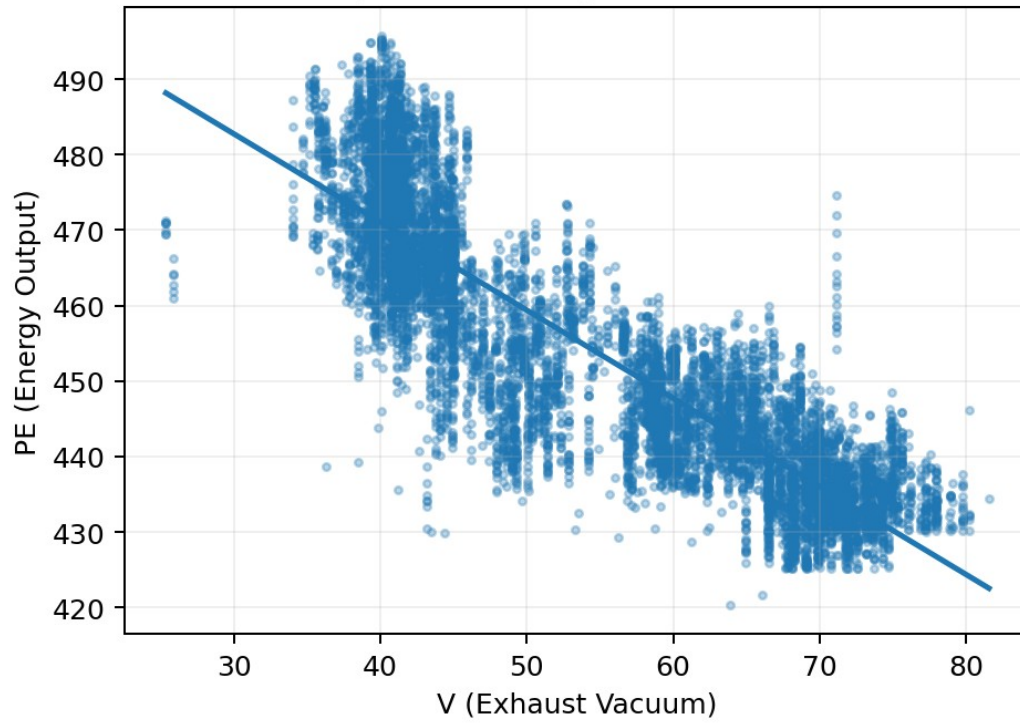
Predictor	Intercept	Slope	p-value	R^2	studentized resid > 3	Max Cook D
AT	497.0341	-2.1713	< 1e-300	0.8989	42	0.0143
V	517.8015	-1.1681	< 1e-300	0.7565	33	0.0032
AP	-1055.2610	1.4899	< 1e-300	0.2688	29	0.0082
RH	420.9618	0.4557	< 1e-300	0.1519	2	0.0021

All four predictors are individually statistically significant at conventional levels. AT and V have strong negative marginal effects on PE; AP and RH have positive marginal effects. AT alone explains about 89.9% of the variance, while V explains about 75.7%.

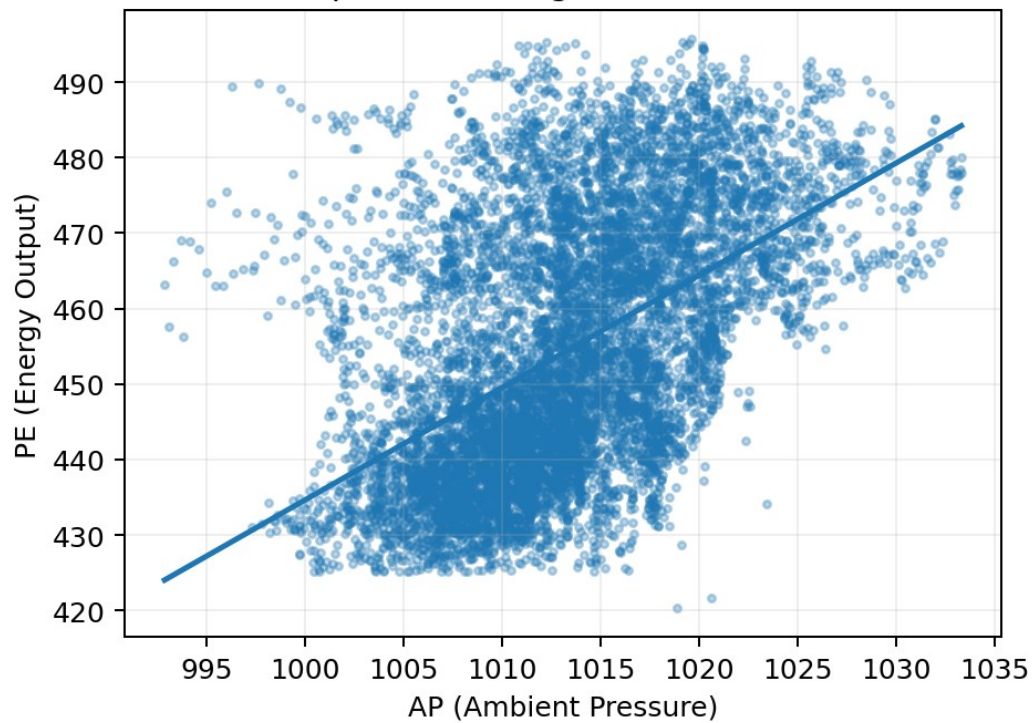
Outlier/influence assessment: counts above use internally studentized residuals. All maximum Cook's distances are below 0.015, but small Cook's distances do not establish that no outliers exist. These diagnostics flag observations for review, not automatic deletion. With no evidence of measurement errors, all observations are retained in each regression.



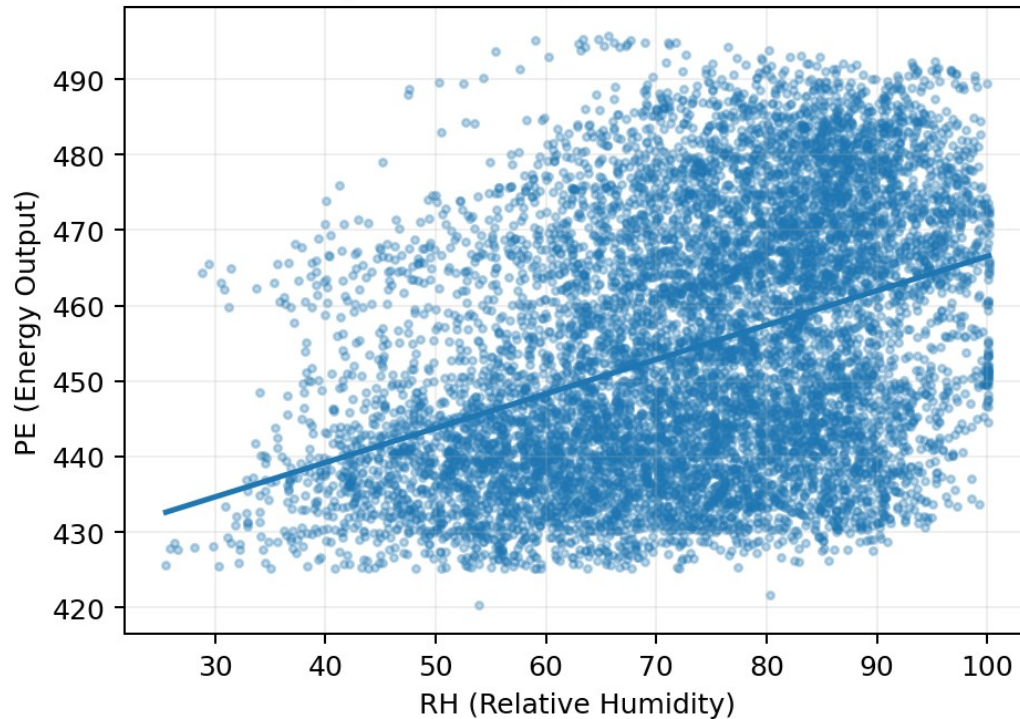
Simple Linear Regression: PE on V



Simple Linear Regression: PE on AP



Simple Linear Regression: PE on RH



(d) Multiple linear regression

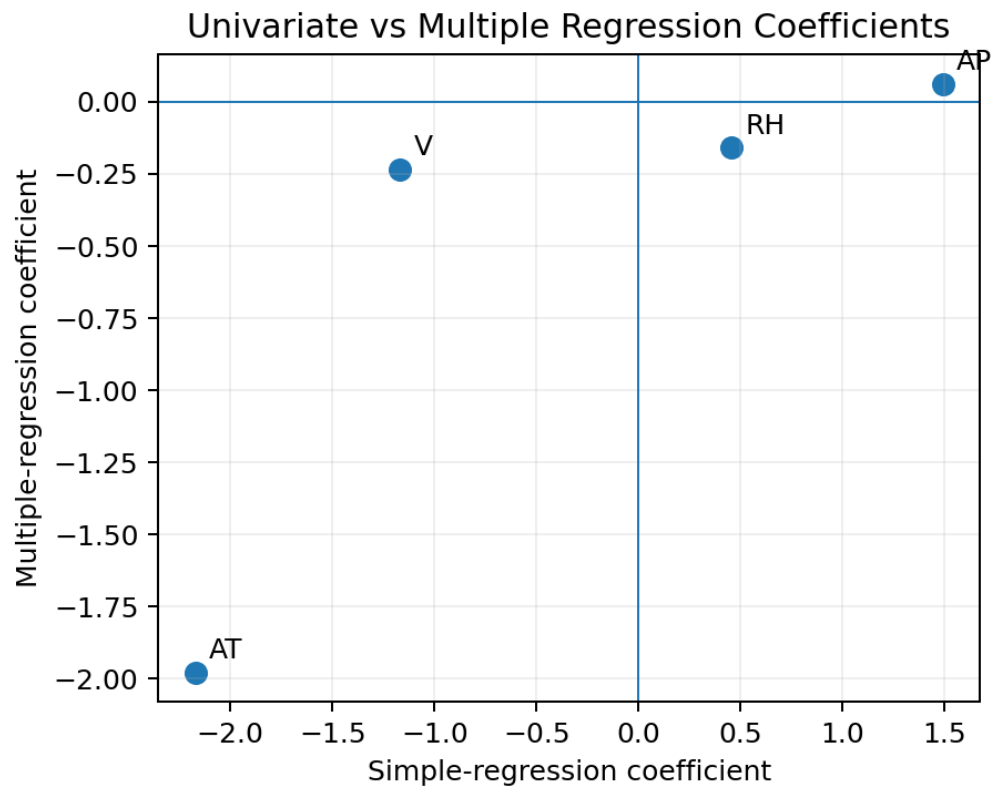
Term	Coefficient	p-value
Intercept	454.609274	< 1e-300
AT	-1.977513	< 1e-300
V	-0.233916	4.38e-215
AP	0.062083	5.51e-11
RH	-0.158054	3.10e-293

The model has $R^2 = 0.9287$. We reject $H_0: \beta_j = 0$ for AT, V, AP, and RH (all $p < 0.05$). The fitted model is $PE = 454.6093 - 1.9775AT - 0.2339V + 0.0621AP - 0.1581RH$.

(e) Comparing simple and multiple-regression coefficients

Predictor	Simple coefficient	Multiple coefficient
AT	-2.17132	-1.97751
V	-1.16814	-0.23392
AP	1.48987	0.06208
RH	0.45565	-0.15805

The AT coefficient changes only modestly and remains strongly negative. V becomes much smaller in magnitude after adjustment for the other predictors. AP also shrinks substantially. RH is the most striking case: its simple-regression coefficient is positive, but its adjusted coefficient is negative. This sign reversal is consistent with confounding/multicollinearity among the environmental variables, especially the strong AT-V relationship and AT/RH association.



(f) Nonlinear associations

Predictor	$p(z^2)$	$p(z^3)$	Joint p for added nonlinear terms	Cubic-model R^2
AT	3.31e-231	3.65e-110	3.48e-285	0.9119
V	8.98e-131	0.0137	7.04e-165	0.7750
AP	2.19e-46	2.12e-67	4.21e-84	0.2975
RH	0.1014	1.44e-05	3.80e-05	0.1537

For numerical stability, each predictor was centered and scaled before forming squared and cubic powers. There is strong evidence of nonlinear association for all four predictors based on the joint comparison of the cubic model against the corresponding linear model. AT, V, and AP have individually significant quadratic and cubic terms. For RH, the squared term is not individually significant (p about 0.10), but the cubic term and the joint nonlinear test are significant; hierarchical modeling still keeps lower-order terms when a higher-order term is retained.

(g) Pairwise predictor interactions

Term	Coefficient	p-value	Significant at 0.05?
AT	-4.347014	0.0670	No
V	-7.674858	1.37e-08	Yes
AP	-0.152355	0.0474	Yes
RH	1.570907	0.0423	Yes
AT:V	0.020971	3.33e-117	Yes
AT:AP	0.001759	0.4521	No
AT:RH	-0.005230	1.22e-10	Yes
V:AP	0.006812	2.88e-07	Yes
V:RH	0.000839	0.0862	No
AP:RH	-0.001612	0.0336	Yes

Significant pairwise interactions are: AT:V, AT:RH, V:AP, AP:RH. AT:AP and V:RH are not significant at the 0.05 level in this full interaction model. Main effects are retained regardless of an individual p -value when they participate in retained interaction terms.

(h) Linear model versus selected quadratic/interaction model

I used one reproducible random 70/30 split (random seed 42). Model 1 contains the four main effects only. Model 2 started with all four main effects, all four quadratic terms, and all six pairwise interactions. Backward elimination removed terms with $p > 0.05$ while respecting hierarchy: a main effect was not eligible for deletion while a retained square or interaction depended on it.

Model	Train MSE	Test MSE
All-predictor linear model	20.5808	21.2399
Selected quadratic + interactions	17.8908	18.6600

Removed terms (in elimination order): V:RH ($p=0.867$), V^2 ($p=0.608$), V:AP ($p=0.358$).

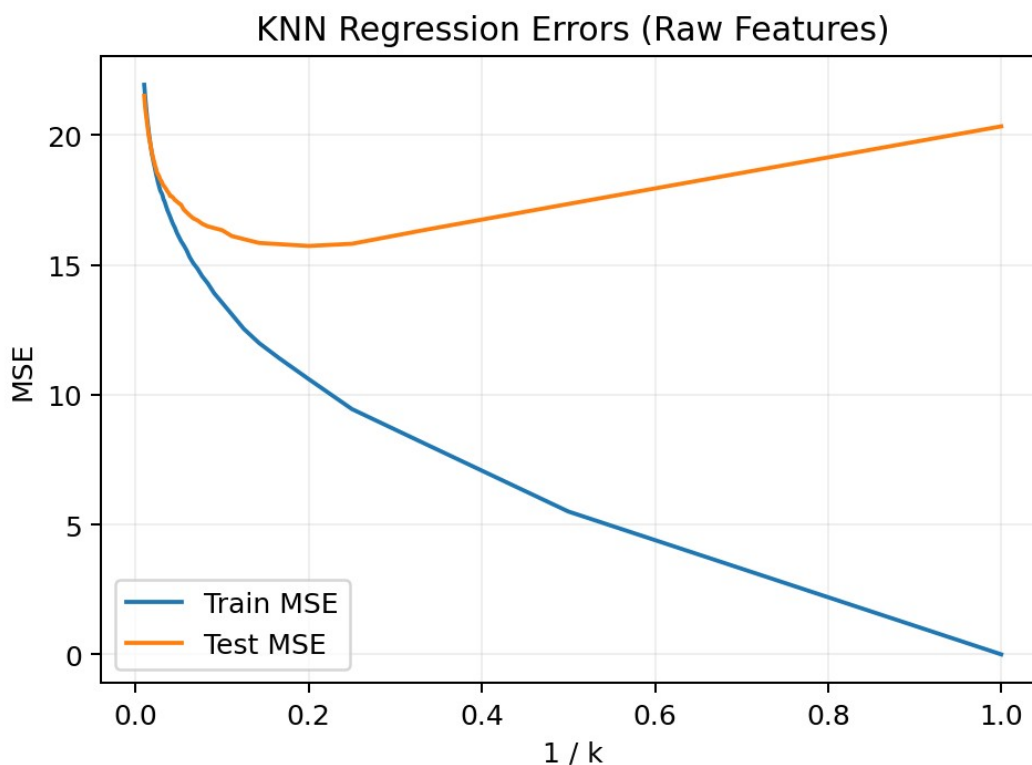
Retained terms: AT, V, AP, RH, AT^2 , AP^2 , RH^2 , AT:V, AT:AP, AT:RH, AP:RH.

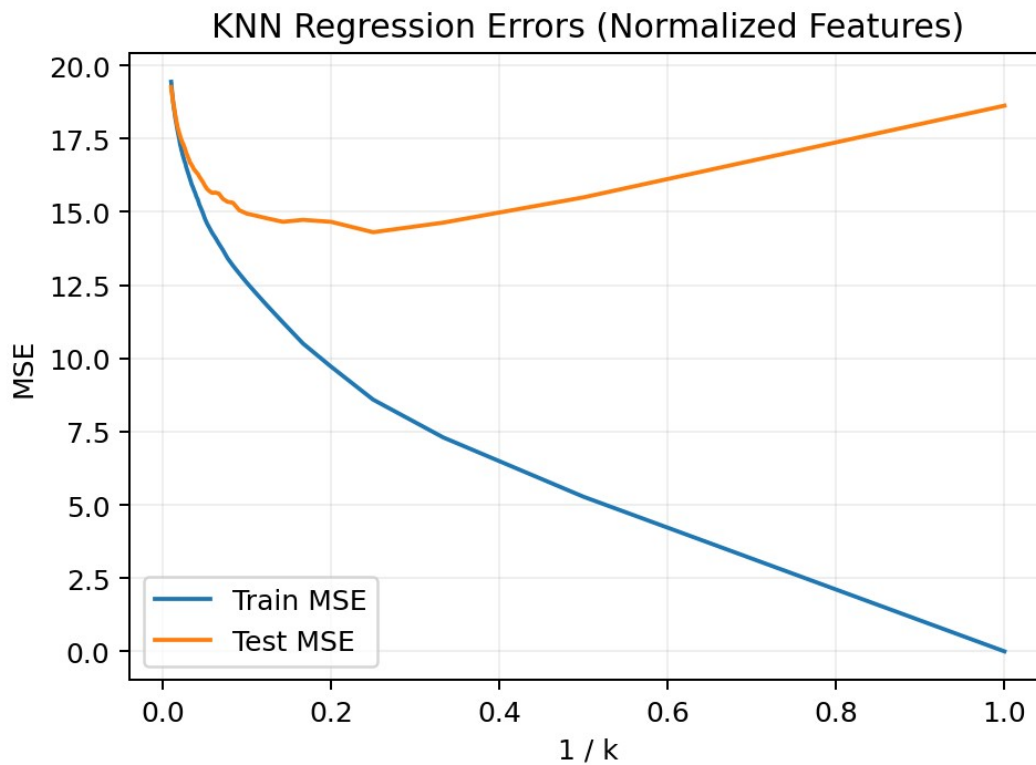
The selected nonlinear/interaction model lowers test MSE from 21.2399 to 18.6600, an improvement of 12.1%. This supports the conclusion that modest nonlinearities and interactions improve the purely linear specification.

(i) KNN regression

Using the same training/test split, I evaluated $k = 1, \dots, 100$ for raw features and for standardized features. Standardization parameters were estimated from the training set only.

Feature version	Best k	Train MSE at best k	Test MSE at best k
Raw	5	10.6008	15.7268
Normalized	4	8.5914	14.3057





Standardization improves KNN on this split because distance depends on differences in feature values. Different units and spreads change their relative contributions. AP's mean near 1000 does not itself increase distances. Scaling uses training-set statistics only.

(j) KNN versus the best linear-regression-family model

The best regression model from part (h) has test MSE 18.6600. The best raw-feature KNN has test MSE 15.7268, and normalized KNN has the smallest test MSE at 14.3057 with $k = 4$. Thus normalized KNN reduces test MSE by about 23.3% relative to the selected quadratic/interaction regression. The result is consistent with KNN capturing local nonlinear structure that is not fully represented by a global low-degree polynomial. The tradeoff is interpretability: the regression coefficients provide direct effect summaries, while KNN mainly targets predictive accuracy.

Evaluation note: k is selected by minimizing the plotted test MSE for this exercise. This minimum is a model-selection result, not an independent final evaluation. A separate validation set or cross-validation would be appropriate for a new predictive study.

2. ISLR 2.4.1

- (a) Better. When n is extremely large and p is small, a flexible method has enough observations to estimate a more complex relationship with less danger of overfitting.
- (b) Worse. When p is extremely large relative to n , a flexible method can fit noise and has high variance, so an inflexible method will generally generalize better.
- (c) Better. If the true predictor-response relationship is highly nonlinear, flexibility is needed to approximate that shape; an inflexible method is more likely to be biased.
- (d) Worse. When the irreducible error variance is very high, a flexible method can chase noise, increasing variance without reducing irreducible error.

3. ISLR 2.4.7

The six training observations are (0,3,0, Red), (2,0,0, Red), (0,1,3, Red), (0,1,2, Green), (-1,0,1, Green), and (1,1,1, Red). The test point is (0,0,0).

Observation	Distance to (0,0,0)	Class
1	3.0000	Red
2	2.0000	Red
3	3.1623	Red
4	2.2361	Green
5	1.4142	Green
6	1.7321	Red

(a) The distances are 3, 2, $\sqrt{10}$ approximately 3.1623, $\sqrt{5}$ approximately 2.2361, $\sqrt{2}$ approximately 1.4142, and $\sqrt{3}$ approximately 1.7321.

(b) For $K = 1$, the prediction is Green because observation 5 is the single nearest neighbor.

(c) For $K = 3$, the three nearest neighbors are observations 5 (Green), 6 (Red), and 2 (Red), so the majority vote is Red.

(d) If the Bayes decision boundary is highly nonlinear, we generally expect a smaller K to work better because small K produces a more flexible, locally adaptive boundary. A large K oversmooths the boundary and can miss nonlinear detail.

References

P. Tufekci, "Prediction of full load electrical power output of a base load operated combined cycle power plant using machine learning methods," International Journal of Electrical Power & Energy Systems, vol. 60, pp. 126-140, 2014.

H. Kaya, P. Tufekci, and S. F. Gurgun, "Local and Global Learning Methods for Predicting Power of a Combined Gas & Steam Turbine," Proc. ICETCEE, 2012, pp. 13-18.

G. James, D. Witten, T. Hastie, and R. Tibshirani, An Introduction to Statistical Learning, Chapter 2 exercises 1 and 7.